METRONIDAZOLE

No alcohol (48 h after)

Metallic taste

CHEMICAL NAME 2-(2-methyl-5-nitro-1H-imidazol-1-yl)ethanol
CLASS Nitroimidazole antibacterial and antiprotozoal
TRADE NAMES Flagyl · Metrogyl · Rozex (gel) · Metronid

THERAPEUTIC USES / INDICATIONS

- Anaerobic infections: intra-abdominal, pelvic, dental, aspiration pneumonia, brain abscess.
- *C. difficile* colitis (mild, where vancomycin unavailable).
- Protozoa: amoebiasis (dysentery, liver abscess), giardiasis, trichomoniasis.
- Bacterial vaginosis; *H. pylori* eradication; surgical prophylaxis (colorectal).

ADMINISTRATION

- **PO** tablets with or after food; suspension (benzoate) 1 h before food.
- **IV** infusion 500 mg over 20–60 min (ready-to-use bag).
- **PR** suppositories; **vaginal** gel; **topical** gel (rosacea).

DOSAGES

- **Anaerobic infection:** 500 mg PO/IV every 8 h.
- **Amoebic dysentery:** 400–800 mg every 8 h for 5–10 days (follow with luminal agent).
- **Trichomoniasis:** 2 g single dose (treat partner) or 500 mg every 12 h for 7 days.
- **Bacterial vaginosis:** 400–500 mg every 12 h for 7 days.
- **Children:** 7.5 mg/kg every 8 h.

PHARMACOKINETICS

- **Peak:** PO 1–2 h. Bioavailability ~100%.
- Excellent penetration incl. CSF, abscesses, bone.
- Metabolised in liver; excreted in urine (may darken urine). **t½:** 8 h.

PHARMACODYNAMICS

- Reduced inside anaerobic cells to toxic free radicals → DNA strand breakage → bactericidal.
- Active against obligate anaerobes and protozoa; no aerobic activity.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Metallic taste, nausea, anorexia.
- Dark (red-brown) urine; harmless.
- Furred tongue, headache.

ADVERSE EFFECTS (SERIOUS)

- Disulfiram-like reaction with alcohol: flushing, vomiting, tachycardia.
- Peripheral neuropathy, seizures, encephalopathy (high/prolonged doses).
- Leucopenia.
- Hepatotoxicity (especially Cockayne syndrome).

CONTRAINDICATIONS & CAUTIONS

- Nitroimidazole hypersensitivity.
- Alcohol use during and 48 h after treatment.
- Use with disulfiram in last 2 weeks.
- **Caution:** first trimester (avoid high single dose), hepatic impairment, CNS disease, blood dyscrasias.

KEY INTERACTIONS

- **Alcohol:** disulfiram-like reaction.
- **Warfarin:** ↑↑ INR.
- **Lithium, busulfan, phenytoin:** ↑ levels/toxicity.
- **Phenobarbital, rifampicin:** ↓ metronidazole.

NURSING MANAGEMENT

- Check alcohol use; explain avoidance.
- Monitor INR if on warfarin.
- Monitor for numbness/tingling, seizures in long courses.
- Monitor CBC with prolonged treatment.
- Treat sexual partners for trichomoniasis.

PATIENT EDUCATION

- No alcohol during treatment and for 48 h after (incl. mouthwash, syrups).
- Metallic taste and dark urine are expected.
- Report numbness, tingling or fits.
- Take with food to reduce nausea.

KEY POINTS

- Covers anaerobes and protozoa.
- Raises INR with warfarin.
- No alcohol — disulfiram-like reaction.
- Neuropathy with long courses.